Preference Elicitation for a Movie Recommender

Human-Centric Artificial Intelligence · Group 29 · Project 4

Introduction

This report covers the method behind a movie recommender that adapts to a new user, and the design of a user study comparing two ways of eliciting that user's preferences. The utility of a film to a user is taken to be linear, $U(x) = w^T x$, and the purpose of elicitation is to estimate w from a small number of interactions. Section 1 sets out the feature representation, section 2 the preference model and its estimator, and section 3 the study. The study is designed, not conducted.

Everything reported here is produced by the implementation in **project4/ml/**, over the 4224 films of the IMDB 5000 Movie Dataset that carry every field the representation needs.

1 Feature representation

1.1 The constraint that decides the design

Two sentences of the specification fix the problem, and they pull against each other. Because U is linear, whatever the representation does not carry cannot be learned at all: a taste for a particular director exists only if some component of x expresses it, which argues for more features. Because w must be estimated from roughly ten interactions, every extra component is another number to identify from the same evidence, which argues for fewer. A representation with one column per director would have thousands of components and none of them would be estimable.

The representation is therefore deliberately small: **22 components**, each chosen because it separates films in a way a person would recognise as taste.

1.2 What is included

Genre	Multi-hot over the 15 most common genres: Action, Adventure, Animation, Biography, Comedy, Crime, Dr
Era	Scaled release year. People have era preferences, and one scaled number captures the monotone part a
Runtime	Scaled duration. A ninety-minute film and a three-hour film are different propositions on a weeknight.
Critical standing	Scaled IMDB score. Separates viewers who follow acclaim from those who do not.
Popularity	Scaled log vote count. Mainstream against obscure. Logged first, because vote counts span five orders of
Production scale	Scaled log budget. A distinct axis from popularity: expensive films can flop and cheap ones can be widely
Director prominence	Scaled log director followers. A cheap proxy for the auteur axis that avoids a column per director.
Family-friendly	Binary, G or PG against the rest. One-hot over the dozen content ratings would spend a dozen componen

1.3 What is excluded

Director and cast identity, plot keywords and country of production are all high-cardinality, and the signal in them worth keeping is already carried by director prominence and popularity. Gross earnings are excluded because they are largely determined by budget and popularity, both of which are present; a third correlated column adds collinearity, and collinear features are precisely what makes w hard to identify from few observations.

1.4 Extraction and scaling

Rows missing any required field are dropped and duplicates on title and year removed. Every continuous column is then standardised. This matters because utility is an inner product: a column's scale sets the scale of its weight, so without standardisation the budget column, of order 10^8, and the genre columns, which are zero or one, could not share a sensible prior and the regularisation would penalise them wildly unevenly.

2 Preference model

2.1 From a comparison to a ranking

Preferences are assumed to follow a Bradley–Terry model, which gives the probability that one film is preferred to another from their utilities:

$$P(i > j) = exp(U_i) \,/\, [\; exp(U_i) + exp(U_j) \;]$$

One of the two interfaces asks a participant to rank ten films, so the model has to describe a whole ordering rather than a single comparison. The extension used is the **Plackett–Luce model**, which treats a ranking as produced sequentially: the participant picks their favourite from the set, then their favourite of what remains, and so on, each pick following the Luce choice rule.

$$P(i_1 > i_2 > \ldots > i_n) = \Pi_k \; exp(U_{i_k}) \,/\, \Sigma_{j\geq k} \; exp(U_{i_j})$$

2.2 Why this extension

First, it genuinely contains the model we started from. At n = 2 the product has a single factor, $\exp(U_1)$ / $[\exp(U_1) + \exp(U_2)]$, which is exactly Bradley–Terry. Both interfaces are therefore described by one model with one w, and their results are directly comparable — which is the entire point of the study. An extension that did not reduce would be a second model in disguise, and the comparison would be confounded by the choice of model rather than the choice of interface.

Second, the generative story it encodes is a plausible account of what a person actually does when ranking: choose the best, set it aside, choose the best of the rest.

2.3 The alternative that was rejected

A ranking of ten implies 45 pairwise comparisons, and these could be fed to Bradley–Terry as if they were separate observations. This needs no new model and is simpler. It is rejected because those 45 comparisons are not independent: they arise from one ordering produced by one person in one act. Treating them as independent multiplies the same evidence many times over, so the likelihood is misspecified and the apparent precision of w is badly overstated. In a study whose purpose is to compare how much each interface reveals about w, an estimator that inflates its own confidence for one of the two designs would decide the outcome before any data was collected.

2.4 Estimating w

The log-likelihood of a set of observed rankings is concave in w. It is maximised under a Gaussian prior:

$$w\blacksquare = argmax \quad \Sigma_r \; log \; P(ranking_r \mid w) \;-\; (\alpha/2) \; ||w||^2, \quad \alpha = 1.0$$

The prior is not decoration. With roughly ten interactions and 22 features the unpenalised maximum is not unique: any direction the shown films do not distinguish is unconstrained, and the optimiser will run off along it. The penalty holds those directions at zero, which is the honest statement that the data said nothing about them. Optimisation uses L-BFGS with the analytic gradient.

Two properties were checked against the implementation. At n = 2 the Plackett–Luce probability equals the Bradley–Terry probability to 10^{-12}. The analytic gradient agrees with central differences to 10^{-8}.

3 User study

3.1 Research question and hypothesis

The study compares two elicitation interfaces. In Design 1 the participant is shown two films and chooses the one they would rather watch. In Design 2 they are shown ten films and rank them.

The two are not equally expensive, and this is the crux of the design. A pairwise choice yields at most one bit and takes seconds. Ranking ten yields up to $\log_2(10!) \approx 21.8$ bits but takes considerably longer and asks more of the participant. So the question ‘which elicits w better’ is under-specified: **better per what?** Per interaction, ranking wins almost by construction and the study would be pointless. Per minute of participant time it is a genuine question. Fixing the budget by which the comparison is normalised is the single most consequential decision in this design.

H1. For a fixed amount of participant time, ranking ten films yields a more accurate estimate of w than repeated pairwise choices.
H0. There is no difference in accuracy per unit of participant time.

3.2 Design

Type	Experimental. The interface is manipulated directly and assignment is controlled, which permits a causal c
Assignment	Within-subjects: every participant uses both interfaces.
Why within	Taste varies enormously between people, and that variance is far larger than the expected effect of the in
Order control	Counterbalanced, alternating across participants. Within-subjects makes order a confound, since whichev
Other controls	Practice trials before each block, and a break between blocks. Films are drawn uniformly at random from t
Trials	Eight per interface; ranking sets of ten films.
Validation	Six further pairwise choices at the end, never used to fit w. They are what makes the primary measure co

3.3 Measures

- **Primary.** The share of the held-out choices that a preference vector fitted from one interface alone predicts correctly, divided by the minutes that interface consumed. Both interfaces are scored against the same held-out block, so they are compared on equal terms.
- **Secondary.** Time per trial; the remaining uncertainty about w; agreement of the resulting top-k recommendations between the two designs.
- **Subjective.** Perceived effort per interface, and which the participant felt described their taste better, both from the closing questionnaire.

Response time is recorded per trial in the browser rather than on the server, so that it measures the participant's thinking time and not network latency.

3.4 Participants and recruitment

- Any adult who watches films; no domain expertise is required, since the task is about personal taste rather than knowledge.
- Recruited through university mailing lists and noticeboards, with the study run online so participants take part at a time that suits them.

- Compensation offered for their time, since asking for unpaid participation is both an ethical problem and a source of self-selection bias.
- Sample size determined from a pilot before recruitment begins, rather than asserted; the pilot also provides the variance estimate needed to do so.

3.5 Procedure

Informed consent, before anything is recorded. Then instructions for the first interface, practice trials, the first block of eight, a break, instructions for the second interface, practice, the second block, the questionnaire, and a debrief explaining what was being compared and how to request erasure. Piloting with five to ten participants precedes the study proper, in person where possible so that confusion can be observed directly; pilot data is excluded from the final analysis.

3.6 Ethics and data protection

- Informed consent is required before any data is recorded, and the interface cannot be started without it.
- Only a pseudonymous random identifier is stored. No name, no email address, no IP address.
- Participation is voluntary and can be ended at any point, without giving a reason and without penalty.
- Data minimisation: only the responses and their timings, because both are required by the stated measures and nothing else is.
- Right to erasure. The identifier is shown at the end so a participant can request deletion; it is the only handle linking their responses together.
- Review by the university ethics board before recruitment begins.

3.7 Analysis plan

Exclusion criteria are fixed before collection, not after: trials whose response time lies more than three standard deviations from the mean, participants whose answers show no variation, and anyone failing the attention check. Deciding these afterwards would allow the data to be filtered towards the hypothesis.

Analysis then proceeds from descriptive statistics per condition, to a paired test on the primary measure — paired because the design is within-subjects and each participant provides both conditions — to the secondary measures with a correction for multiple comparisons, since testing several outcomes at $\alpha = 0.05$ without one would make a spurious result likely.

4 Pilot on simulated participants

Lecture 7 frames evaluation in this field as two steps: simulated users first, because they give full control over behaviour and real users are costly, then human users. The study above is step two. This is step one, and it settles two things argument cannot: whether the estimator recovers a preference vector at all from the interactions the study can afford, and how many participants the real study needs.

250 simulated participants per time budget, each with a known w, answering by the Plackett-Luce model itself with beta = 3.0 -- someone who mostly follows their own taste without being mechanical. Both designs receive the same time budget and spend it on as many trials as they can afford, at an assumed 6s per pairwise choice and 45s per ranking of ten.

Budget	Trials	Pairwise	Ranking	Ceiling	Diff	dz	n
2 min	20/2	70.0%	70.9%	80.5%	+0.9%	+0.09	1066
4 min	40/5	72.8%	74.8%	79.6%	+2.0%	+0.22	166
6 min	60/8	73.5%	77.0%	80.6%	+3.5%	+0.43	45
8 min	80/10	75.0%	76.8%	79.6%	+1.8%	+0.25	127

4.1 Reading the result

Agreement has to be read against the ceiling rather than against 100%. A participant who is not perfectly consistent with themselves caps how well any preference vector can predict their held-out choices, including their own true one; that ceiling sits near 80%, so a design reaching 77% has recovered most of what was recoverable.

Ranking leads at every budget, but only barely at the shortest. At two minutes a ranking block affords two trials, too few to constrain 22 weights, and the advantage nearly vanishes. This is a design finding in its own right: too short a session does not merely weaken the study, it erases the effect it is trying to measure.

The effect is small. Where ranking leads it leads by one to three percentage points, and the sample sizes that follow range from about 45 to over a thousand. That spread is itself informative: with an effect this small, even 250 simulated participants cannot pin the required n tightly. The defensible reading is that the study needs on the order of one to two hundred participants rather than the twenty a coursework study would otherwise assume, and that quoting the most flattering row would be a mistake.

4.2 What the timing assumption is carrying

The comparison is normalised by time, and a simulation cannot know how long a trial takes, so the durations are assumptions. Varying the assumed time for a ranking shows the direction holds while a ranking takes up to about a minute and disappears at ninety seconds, where only four rankings fit the budget. The conclusion therefore depends on the assumption, but not delicately: it fails only in a regime where the ranking block is too short to be informative anyway. The interface records real per-trial times, so a human pilot replaces this assumption with measurement.

None of this is evidence about people. Every participant here follows the model exactly, which is the one thing a real participant certainly does not do. The pilot establishes that the instrument works and what sample the study needs; it cannot establish that the hypothesis is true.

The interface described in section 3 is implemented and reachable from the project landing page. This report is generated from the implementation by **manage.py build_project4_report**, so the feature list, dataset size and prior strength quoted above cannot drift from the code.